

# Chapter 11

## AI in Substance Use and Addiction Prevention



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### 1 Introduction

Substance use disorder (SUD) represents a critical and entrenched social welfare challenge, with nearly 50 million individuals meeting criteria in the United States (US) (Substance Abuse and Mental Health Services Administration [SAMHSA], 2024). Driven by complex biological, psychological, and social factors, SUD can be especially difficult to assess, prevent, and treat among marginalized communities (Leibowitz et al., 2023; Schoenfeld et al., 2019; Skewes & Gonzalez, 2013). Artificial intelligence (AI), with its rapidly advancing role in healthcare, has the potential to transform how social workers, allied professionals, and community members understand and address SUD (Poalelungi et al., 2023; Goldkind, 2021).

AI is a broad umbrella term that refers to computer and software systems built to mimic human behavior or solve complex problems that would otherwise require human intelligence (Russell & Norvig, 2021). Many contemporary AI applications for SUD rely on machine learning, a method that combines statistical techniques with computer algorithms. Machine learning can detect patterns in data and make predictions, such as whether a client is at risk of a SUD, or assist with decision-making, such as selecting an optimal evidence-based treatment (Suva & Bhatia, 2024). There has also been growth in the use of deep learning, an advanced form of machine learning that uses layers of connected “neurons” (similar to how the human brain works) to automatically uncover patterns in increasingly complex data like brain scans or audio recordings from client encounters (LeCun et al., 2015).

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In this chapter, we aim to explore both AI's promise and looming challenges for addressing the problematic or harmful use of psychoactive substances, including alcohol, opioids, cannabis, stimulants, and other drugs, and we consider the implications of AI for vulnerable substance using populations. For context, we first provide background information on SUD and traditional treatment and prevention strategies. Then, we review advances in AI and their applications to SUD, explore ethical considerations of AI and the importance of engaging stakeholders and clients in AI development, and consider the future of this field.

Below, we examine how AI is being used for risk assessment, treatment, and relapse prevention, and focus especially on prevention, early intervention, and harm reduction to align with contemporary best practices in social work and public health that emphasize making available a continuum of strategies to meet individuals where they are (Paquette et al., 2022; Stallvik et al., 2015; Vakharia & Little, 2017). As we demonstrate, while most work on AI for SUD has been limited to research settings, AI is beginning to impact real-world practice and shows compelling potential. At this juncture, it is imperative that ethics not take a back seat to technological advancement. In line with social work's core values of promoting human dignity, social justice, and community well-being (National Association of Social Workers [NASW], 2021), we highlight how AI, when ethically implemented, can help make SUD treatment and prevention practices more equitable, accessible, and relevant for vulnerable populations.

## 2 Contextualizing Substance Use Disorders

### 2.1 *SUD Definitions, Epidemiology, and Social Welfare Burden*

SUD is a diagnostic classification with criteria set by the American Psychiatric Association's Diagnostic and Statistical Manual of Mental Disorders (the DSM-5; American Psychiatric Association [APA], 2013), which is used by US mental health practitioners, including licensed clinical social workers, to identify clinical disorders. To meet criteria for SUD, a client must demonstrate symptoms related to the use of substances (e.g., taking more of a substance than initially intended, challenges controlling use, cravings, tolerance, and withdrawal symptoms), along with health and social consequences of use, and continued use despite these consequences (APA, 2013). Within the DSM-5, there are separate SUD diagnoses for each specific substance (e.g., *alcohol* use disorder for SUD involving alcohol, *opioid* use disorder for SUD involving opioids, etc.).

There are several other terms used in clinical and research settings to refer to the problematic use of substances, some of which are at times used interchangeably with "SUD," including "dependence," "addiction," "misuse," "abuse," "heavy use," "risky use," and "problematic use" (Rehm et al., 2013). Though a full discussion of

these terms is beyond the scope of this chapter, unless otherwise specified, we will refer to “SUD” when discussing problematic substance use that meets criteria for a clinical disorder and “problematic substance use” as a catch-all umbrella phrase that includes SUD as well as less severe symptom presentations.

SUD is a widespread, deeply entrenched, and challenging social welfare problem. Over 2.2% of the world’s population meets criteria for SUD, with global prevalence steadily rising over the past several decades (Castaldelli-Maia & Bhugra, 2022; Pan et al., 2020). In the United States, over 48 million individuals had SUD in 2023, and rates have more than doubled over the past 10 years (SAMHSA, 2024). Alcohol has consistently been the primary driver of SUD, with over half of cases involving alcohol, followed by cannabis and opioids (SAMHSA, 2024). An increasing prevalence of cannabis use disorder in the United States (SAMHSA, 2024), including among young people (Han et al., 2017), may be due in part to the changing legal status of cannabis in recent years. The opioid epidemic, involving high rates of opioid use disorder and deaths due to overdose, represents one of the most difficult public health challenges the United States has faced in recent years (Skolnick, 2018). Emerging adults (18–25 years old) have had the highest rates of nonmedical use of prescription opioids (NUPO) and, compared with older age groups, may experience barriers to effective care (Tapscott & Schepis, 2013; Schuman-Olivier et al., 2014). Encouragingly, recent data indicate a downward trend in NUPO, and improvements in treatment and prevention (such as through technology and AI) could further ameliorate the problem (McCabe et al., 2025; Saxon et al., 2025).

SUDs impose a significant societal burden, resulting in billions of dollars in annual costs due to lost productivity and increased utilization of the healthcare and criminal justice systems (Manthey et al., 2021). The adverse effects of SUD on individuals, families, and communities include hindering physical and mental health, contributing to early mortality, and disrupting interpersonal and social functioning (Newcomb & Locke, 2005). Critically for social workers, many vulnerable and marginalized populations are at heightened risk of SUD, including individuals experiencing poverty (Baptiste-Roberts & Hossain, 2018), sexual and gender minorities (Connolly & Gilchrist, 2020; Hughto et al., 2021), American Indian/Alaska Native people (Soto et al., 2022), trauma survivors (Levin et al., 2021), and those with a history of incarceration (Fazel et al., 2017) or homelessness (Gutwinski et al., 2021). These individuals are, in turn, vulnerable to experiencing stigma, criminalization, and barriers to necessary healthcare because of their substance use (Newcomb & Locke, 2005).

## ***2.2 Biopsychosocial Determinants of SUD***

To understand the persistence of SUD as a social welfare problem and to inform effective intervention and prevention, it is critical to have an accurate grasp of its etiology and the factors that contribute to SUD. Problematic substance use is often

progressive in terms of severity, with individuals transitioning from substance use initiation to greater levels of consumption, then developing mild and then more severe SUD (Lee et al., 2019). In terms of *why* some individuals develop SUD, traditional biomedical models emphasize physiological causes, including a genetic predisposition and substance-related changes to the brain (Skewes & Gonzalez, 2013). In contrast, a contemporary, biopsychosocial understanding of SUD espouses that psychological, sociocultural, *and* biological factors interact in complex ways to confer risk for the development, maintenance, and recurrence of SUD (Skewes & Gonzalez, 2013). Notably and in line with biopsychosocial models of health, 80% of health outcomes are explained by social determinants rather than clinical or biological factors (Greer et al., 2023). Social workers have a substantial role to play in integrated SUD care where multiple disciplines collaborate to address these risk factors holistically (Kaur & Maleckas, 2025), given the profession's focus on social functioning and training in biopsychosocial models of health.

There is a robust empirical literature substantiating the biopsychosocial view of SUD. Recent research confirms there are genetic risk factors for SUD (Hancock et al., 2018) and chronic substance use reshapes reward and stress circuits in the brain (Volkow et al., 2016; Koob & Volkow, 2016), but these changes only partially explain SUD risk. A breadth of psychological and sociocultural risk factors for SUD have been identified, ranging from trauma exposure and mental health conditions to peer and family influences and neighborhood disadvantage (Belfiore et al., 2024). Researchers have also demonstrated how biological and social factors *interact*, for instance, via large-scale adoption studies showing that familial challenges can lead to worse outcomes among those who are genetically predisposed to SUD (Kendler et al., 2012). Substance use and exposure to risk factors can also have differential influences on the risk of developing SUD, depending on an individual's age and developmental stage (Lee et al., 2019). It is also important to note that macro-level policy influences SUD risk to a large extent, such as via drug availability, criminality, and access to treatment (Strang et al., 2012).

Many biopsychosocial determinants of SUD may especially heighten risk among marginalized populations. These populations often have higher rates of trauma exposure, mental health conditions, and neighborhood disadvantages, which are all associated with SUD (George et al., 2023; Mikhail et al., 2018). Individuals experiencing poverty may lack access to healthcare, including SUD treatment (Olfson et al., 2022), and discrimination in healthcare settings contributes to disparities in treatment quality among racial/ethnic minority individuals (Mays et al., 2017). Minority stress models, developed to characterize the unique psychological and social stressors that sexual minority individuals face, may in part explain substance use in this population (Goldbach et al., 2014). It is also vital to consider how identifying with multiple marginalized identities likely confers unique SUD-related risks, in line with an intersectional perspective (Otiniano Verissimo et al., 2023; Schilt-Solberg et al., 2025).

### ***2.3 Contemporary Treatment and Prevention Approaches***

Contemporary SUD treatment exists on a continuum, ranging from low-intensity, brief interventions or outpatient treatment to more intensive inpatient or residential programs (Mee-Lee et al., 2013; McLellan et al., 2000). Care recommendations are ideally made based on an individual's symptom severity and biopsychosocial risk factors such as co-occurring mental health conditions and social challenges (Mee-Lee et al., 2013; McLellan et al., 2000). Psychological interventions—such as cognitive-behavioral therapy (CBT), motivational interviewing, and contingency management—are commonly used to enhance motivation for change, build coping and recovery skills, and help individuals in remission avoid relapse (Carroll & Onken, 2005; Magill et al., 2019). For individuals with moderate to severe SUD (particularly opioid use disorder), evidence supports medication-assisted treatment (MAT), in which clients receive medications such as buprenorphine or naltrexone to block cravings and prevent misuse (SAMHSA, 2021; Maisel et al., 2013).

Integrated care approaches that combine MAT, psychological interventions, and social support have been associated with improved retention in care, reduced substance use, and fewer relapse events, likely because they more fully address the biological, psychological, and social dimensions of SUD (Leibowitz et al., 2023; National Institute on Drug Abuse [NIDA], 2018; Sharma et al., 2020). Hub and spoke models, implemented in Vermont and nationally, represent an effective systems-based approach to treating opioid use disorder, integrating medical, behavioral, and wellness care (Brooklyn & Sigmon, 2017). Regional hubs include specialty opioid treatment programs providing intensive services, including induction and dosing of medication, and spokes include community-based healthcare settings that support patient self-efficacy and recovery capital (Best & Nisic, 2022). While traditional SUD treatment has focused on immediate cessation of use and abstinence, MAT and hub and spoke programs prioritize a low-barrier approach and easy access to effective treatment. Such programs are consistent with harm reduction principles when they minimize stigma and the negative consequences of substance use, even for those not motivated to pursue abstinence (Hawk et al., 2015).

Strategies to prevent SUD from developing in the first place also exist on a continuum (Volkow & Blanco, 2023). Universal prevention programs aim to decrease risk among an entire population, such as school-based curricula to enhance psychosocial skills or mass media campaigns to reduce early experimentation. Selective prevention programs target specific groups with known biopsychosocial risk factors, such as programs targeting youth in disadvantaged neighborhoods or children of caregivers with SUD. Even more targeted, indicated prevention programs focus on individuals who exhibit early signs of problematic substance use but do not yet meet criteria for SUD, such as through screening and brief intervention for problematic use in a healthcare setting. The most common prevention programs are universal, school-based education programs targeting all children and adolescents, rather than those with specific risks (Faggiano et al., 2014). Relatively successful

programs often target multiple risk and protective factors at once, in line with biopsychosocial models of SUD (Faggiano et al., 2014; Volkow & Blanco, 2023).

Unfortunately, SUD treatment and prevention programs show only modest success, as evidenced by the high and increasing prevalence of SUD (SAMHSA, 2024), challenges engaging individuals in interventions, and often low levels of intervention effectiveness (Farhoudian et al., 2022). Because SUD is a multifactorial biopsychosocial problem (with many potential intersecting risk factors) and it often develops gradually over time, it is difficult to predict who will develop SUD, who will respond well to treatment, and for whom SUD will recur after treatment (Stallvik et al., 2015; McCabe et al., 2018). Once SUD develops and becomes severe, it can be challenging to treat and to maintain recovery (Restrepo-Guzman et al., 2020). A lack of access to care, stigma around substance use, and competing psychosocial demands can serve as barriers to initiating and staying engaged in treatment, particularly for marginalized populations (Farhoudian et al., 2022; Husain et al., 2023). Screening for problematic use is also time and resource-intensive for health and social service systems.

Prevention brings additional challenges. While clinically validated tools are available to identify current problematic use, providers are less equipped to identify those at risk of developing it later. And while preventative interventions primarily target young people, many are at risk for SUD throughout the life course, and there is a need to disseminate prevention strategies for other age groups and settings (Volkow & Blanco, 2023). In both prevention and treatment, progress has been made in considering the range of biopsychosocial risk factors and targeting them across prevention and integrated treatment strategies. However, the field could benefit from a more personalized approach that assesses and addresses individuals' specific needs and preferences in an increasingly holistic and time-sensitive manner (Marchand et al., 2019). AI has the potential to transform SUD treatment and prevention, specifically in this way.

### **3 Emergence of AI in SUD Treatment and Prevention**

#### ***3.1 Historical Overview of AI in Medicine and Behavioral Health***

AI has been applied in medicine and behavioral healthcare throughout its history and evolution. In the mid-1960s, the first chatbot, ELIZA, was designed to mimic psychotherapy by recognizing simple word patterns and generating predefined responses (Weizenbaum, 1966). During this era, medical research databases and health records became increasingly digitized, laying the foundation for future developments in healthcare data science and AI (Kulikowski, 2019). In the 1970s, AI research in medicine continued to emerge with the development of expert systems like MYCIN, which used rule-based reasoning to diagnose and recommend

treatment for bacterial infections (Kaul et al., 2020). Methodological advances in the 1990s and 2000s enabled AI to analyze complex datasets, leading to breakthroughs in medical applications, including in medical imaging and diagnostics, genomics, and personalized and precision medicine (Kaul et al., 2020).

The 2010s saw rapid growth in the use of AI for behavioral health, including substance use research. At that time, researchers began to demonstrate how AI algorithms can be applied to accurately predict risk for mental health conditions, including depression, schizophrenia, and suicide (Abd Rahman et al., 2020). Concurrently, there was renewed interest in developing AI-powered treatment applications, such as improved chatbots, for mental health disorders (Moore & Caudill, 2019); the earliest efforts related to SUD involved using AI algorithms to improve screening for problematic substance use. One early study demonstrated how an AI model could identify heavy drinkers and delineate risk and protective factors for drinking among college students (Zuba et al., 2012); another used AI to categorize individuals based on the characteristics of their opioid use disorder to better determine the influence of genetics (Sun et al., 2012). A 2016 study reported one of the first AI treatment applications for SUD, a chatbot to assess problematic alcohol use among young adults and provide them with education on responsible drinking (Elmasri & Maeder, 2016).

Increasingly, AI has garnered interdisciplinary interest, including among social workers. The Grand Challenges for Social Work include a call to “harness technology for social good,” acknowledging that technologies such as AI, when deployed ethically, can help address challenging social problems in new ways (American Academy of Social Work and Social Welfare, n.d.). In cutting-edge university research laboratories, social work scientists lead teams involving engineers, computer scientists, and informaticists to advance such “social good” applications of AI, including SUD (Tambe & Rice, 2018; Davis et al., 2022). Critically, social work scholars are also centering the voices of vulnerable populations who may be affected by AI and advocating that AI applications are ethical, trustworthy, and unbiased (Goldkind et al., 2025). While some may view AI as primarily within the purview of technical disciplines, social workers have much to contribute through dynamic research methodologies, an orientation towards team science and practice, and critical perspectives on social justice.

### ***3.2 AI Tools and Techniques***

Before reviewing contemporary applications of AI to substance use and SUD, it is necessary to describe and delineate the relevant core methodologies of AI. Early AI systems, including the psychotherapy chatbot ELIZA, relied on “knowledge base” approaches, where experts manually encoded rules for the application to follow (e.g., IF the user’s statement includes the word “sad” THEN ask “What makes you feel sad?”). However, this method was limited because humans often draw on unspoken or hard-to-describe knowledge when solving problems. For example, a

clinician treating SUD may make different care recommendations based on nuanced differences in complex biopsychosocial factors and their combinations.

Machine learning, with its ability to learn directly from data and better capture the complexities of real-world decision-making, represents an advance from “knowledge-based” approaches (Suva & Bhatia, 2024). Today, the most common AI applications for SUD involve machine learning (Ng et al., 2025; Garbin et al., 2023; Ebrahimi et al., 2021). As an advanced type of machine learning, deep learning is a powerful paradigm for making accurate predictions with complex data (LeCun et al., 2015). We note, however, that as methodologies become increasingly complex, it can become harder to understand how predictions are made (known as the “black box” problem) (Sendak et al., 2020). This is problematic in care settings, where clinicians and clients need meaningful, actionable knowledge to trust AI predictions.

The rapid proliferation of data and technology is broadly enabling these advances in AI and its applications to SUD. Increased digitization and the rise of big data ecosystems across a variety of domains (e.g., research, healthcare settings, social datasets, and consumer devices) provide ever-expanding sources of data for the development of machine learning models that rely on them to make predictions. Data sources that have been used to develop these models in the field of SUD are as wide-ranging as electronic medical records, including records of diagnoses, prescriptions, and provider notes (Ellis et al., 2019); data on social networks from social media platforms (Hassanpour et al., 2019); survey studies, epidemiological, and large publicly available datasets (Davis et al., 2022; Oliva et al., 2022); and physiological data, including from brain scans (Yang et al., 2023), wearable sensors (Davis-Martin et al., 2021), and genomic records (Hatoum et al., 2021). The ubiquity of smartphones and Internet access also allows for advances in intervention delivery, including AI-enabled treatments delivered via mobile devices (Brezing & Brixner, 2022).

Some algorithmic and hardware tools are especially well-equipped for specific types of data: natural language processing is a machine learning approach that extracts meaning from text data—such as from provider notes, transcripts of audio recordings, or social media posts—and this approach can be used to uncover risk for SUD in both micro and macro contexts (Singleton et al., 2023). Deep learning models are particularly well-suited for complex multimodal data such as neuroimaging, genomic, or sensor data, from which they can uncover patterns of risk that might be undetectable with traditional analytical methods (Miotto et al., 2018). Consumer wearable devices, such as Apple Watch and FitBit, represent mainstream adoption of wearable sensors and provide an increasingly accessible source of physiological data for continuously monitoring changes in SUD risk and relapse (Oesterle et al., 2022).

### **3.3 *Promise and Potential***

Enabled by the availability of novel data, the development and adoption of AI has the potential to significantly enhance the understanding, prevention, and treatment of SUD among vulnerable populations. Machine learning models have been developed to screen for current problematic use, detect early signs of SUD, predict who is at future risk of SUD, and assess who might respond well to SUD treatment (Ng et al., 2025; Garbin et al., 2023; Ebrahimi et al., 2021). These models, and AI more broadly, could increasingly be leveraged within health systems, treatments, and prevention efforts to help make interventions more personalized, targeted, and better timed. Among those who have already received treatment for SUD, AI tools, including predictive models that leverage wearable device data, may help monitor for relapse risk (Davis-Martin et al., 2021).

Crucially, the proliferation of data and AI enables the development of tools for prevention and treatment that increasingly adhere to the complex biopsychosocial model of SUD. That is, by leveraging and combining multimodal data sources ranging from psychophysiological data to data on local social environments, AI can model risk based on a range of biological, psychological, and social determinants—and their interactions. This is important for working with marginalized populations, for whom it is critical to address the structural and intersecting roots of SUD risk. Digitally delivered, AI-based interventions can also enhance the ability to reach hard-to-reach populations with better-targeted, more accessible treatments and prevention strategies. In a nutshell, AI has the potential to democratize information and intervention and ground SUD strategies in the realities of risk for vulnerable populations.

## **4 Identifying High-Risk Individuals**

### **4.1 *Identifying Current Substance Use Problems***

One promising application of AI in the substance use field is in its ability to identify individuals who are currently exhibiting problematic use. Researchers have demonstrated that machine learning models can be trained on a range of data sources to identify individuals with existing SUD or early signs of SUD using characteristics captured in those data. For example, several studies have leveraged electronic medical record data to train machine learning models, including natural language processing to analyze providers' notes and to screen patients for current SUD with a high degree of accuracy (Ridgway et al., 2021; Blackley et al., 2021). One study demonstrated how machine learning can identify individuals with SUD from their social media profiles (Hassanpour et al., 2019). Using advanced deep learning methods with images and text from users' Instagram profiles, researchers were able to identify individuals with likely alcohol use disorder.

As discussed further below, machine learning and deep learning models hold promise for improving collaborative clinician-client decision-making, which is critical for effectively identifying and addressing SUD. Model performance and trust in machine learning are significantly increased when there is a process for incorporating interdisciplinary stakeholder feedback and perspectives throughout model development. These models have potential real-world applications, for example, assisting health or social service providers in identifying individuals or groups with a high likelihood of an SUD and aiding in diagnosis. In addition, with its ability to mine large and complex sets of data to uncover new associations and interactions, machine learning is being used to identify potential biopsychosocial markers of current SUD, including changes in the brain (Yang et al., 2023), demographic and psychological characteristics (Ahn & Vassileva, 2016), and neurocognitive functioning (Ahn et al., 2016). These findings warrant further validation in clinical research and could lead to the development of novel screening and diagnostic instruments for SUD.

## ***4.2 Predicting Future Substance Use Problems***

Beyond identifying current problematic use, AI is also being used to predict future outcomes related to SUD. A growing body of research shows how machine learning can predict who may go on to develop an SUD, who might suffer specific consequences of use, and who might or might not respond well to treatment. In one exemplary study, a machine learning model was able to predict from someone's electronic health record data whether they would go on to develop alcohol use disorder with 96% accuracy (Ebrahimi et al., 2022). In another study, machine learning models trained on demographic and psychometric data of clients in psychological treatment for alcohol use disorder could predict treatment outcomes as well as trained clinicians (Symons et al., 2019). Researchers have also applied machine learning to predict whether individuals who have been prescribed opioids would go on to develop opioid use disorder (Dong et al., 2021a) or overdose from opioids (Dong et al., 2021b), with promising results. Machine learning models that combine various multimodal data sources might have especially high predictive power. In a recent study of patients with posttraumatic stress disorder, a deep learning model that integrated electronic health record data with data on one's neighborhood socioeconomic conditions demonstrated very good performance for predicting whether the patient would go on to develop SUD over the subsequent 3-month period (Miranda et al., 2024). As these studies exemplify, machine and deep learning can reveal previously unknown relationships and create high-performance, clinically valid predictive algorithms. These, in turn, can inform targeted prevention and treatment efforts.

Predictive models have wide potential applications in clinical and community settings, where they could help identify at-risk individuals before traditional assessments would flag concerns—or among individuals who are not otherwise engaged

in clinical care—and inform early interventions for those most at risk. Just as research on AI-driven screening could help refine markers of current SUD, predictive modeling may also expand knowledge of risk factors for future risk. For example, researchers applied machine learning to data from a longitudinal survey of youth, uncovering 30 lifestyle variables that predicted future SUD (Jing et al., 2020). The nature of these predictors varied with age—shifting from psychological vulnerabilities in late childhood and early adolescence to challenges of socializing in late adolescence and early adulthood—demonstrating how machine learning can enhance nuance in ecological knowledge.

Other innovations include temporal and graphical deep learning-based modeling using large-scale EHR data for predicting opioid use disorder and other medical conditions (Dong et al., 2021a). For example, graph neural networks (GNNs) are a class of deep learning models that analyze data represented as graphs. In these graphs, *nodes* (e.g., individual patients, a treatment provider, and drugs) interact through *edges* (e.g., behavioral patterns and clinical encounters between a patient and a physician or social worker; Wu et al., 2021). Temporal GNNs extend this to capture how graphs evolve, which is critical for predicting future outcomes such as opioid use disorder. While GNNs can predict risk with increasing accuracy, a significant challenge is the need for advanced computing and technical methodologies to analyze the complex relationships embedded in EHR data, including associations between SUD risk and multifaceted predictors such as co-occurring medical conditions, healthcare access, and adverse childhood experiences.

### ***4.3 Effectiveness and Limitations of Risk Identification***

A growing body of research demonstrates the utility of machine learning for identifying and predicting SUD risk. The largest number of machine learning studies exists for SUDs involving alcohol, cannabis, and opioids (see Ebrahimi et al., 2021; Ng et al., 2025; Garbin et al., 2023, for literature reviews of machine learning research involving these substances, respectively). Though results vary, machine learning has demonstrated the ability to identify SUD risk with accuracy exceeding 90%. There is reason to be optimistic that machine learning could significantly improve SUD practice and policy by providing powerful tools for identification and prediction used in healthcare applications.

As machine learning for SUD has primarily been deployed in research contexts to date, its applicability in real-world settings is less clear, and several challenges arise for real-world deployment. Machine learning models are only as useful as the data that is available to train them. As diagnoses and terminology used for clinical substance use problems change over time, models trained on historic EHR data may miss contemporary indicators of SUD. One study that explored a limited set of possible predictors for opioid use disorder in an EHR dataset revealed that the top risk factors were other SUDs, opioid overdose, and heavy opioid use, information that would essentially be “too late” to share with a clinician whose goal was prevention

(Liu et al., 2023). Even when results are promising, the vast majority of machine learning research for SUD has involved training and testing models on a single dataset, and therefore models that have been developed with one sample have mostly not been externally validated with other samples (Ebrahimi et al., 2021; Ng et al., 2025; Garbin et al., 2023). Machine learning is susceptible to overfitting, where it can perform very well on training data but do poorly with new, unseen data because it is fit too specifically to the noise and idiosyncrasies of the training data (Ying, 2019). Thus, it is essential to ensure models trained in one context would perform well in real-world settings before they are broadly deployed.

There is also a large variety in types of machine learning models and in the data that they are trained on (Ebrahimi et al., 2021; Ng et al., 2025; Garbin et al., 2023), making it challenging to compare results across studies and to implement a model that was validated in one setting in another setting. To illustrate, if health system A uses a machine learning model trained on their patients' wearable device data, but health system B does not collect this type of data, a model trained in health system A would be infeasible to deploy in health system B. Many machine learning models also have low interpretability, making it difficult to understand why they make certain predictions or how they would behave under different conditions (Carvalho et al., 2019). This may make it harder for practitioners and stakeholders to trust and apply the model in real-world decision-making, especially when the population, substances, underlying data, or assumptions differ across settings.

## 5 AI-Supported Treatment and Recovery

### 5.1 *AI-Enabled Treatments*

Beyond identifying risk, AI has the potential to improve the delivery of SUD treatment—both by enabling providers to make better care decisions and by developing novel, technology-delivered treatments. Given the complex, biopsychosocial nature of SUD, clients' unique symptom and risk profiles, and providers' time constraints, it can be challenging to make effective clinical decisions and allot resources. Newly developed clinical decision support tools apply predictive models on client data and up-to-date research on best practices to help tailor treatment recommendations to each patient's unique clinical needs (Tai et al., 2024). Early research on these tools demonstrates they could improve SUD outcomes and satisfaction in care (Tai et al., 2024). One can imagine a social worker with a busy caseload benefiting from such a tool by being better informed about high-risk clients and making more timely, informed care decisions for them.

AI is also helping to drive advances in client-facing digital therapeutics, which are treatment or prevention strategies delivered via technology directly to at-risk populations (Hao et al., 2024). Technology is helping expand how care is offered, including via the Internet, mobile devices, and virtual reality, potentially lowering

treatment barriers through telehealth and enabling creative, flexible treatment adaptations for vulnerable populations. Increasingly, AI is being integrated into these systems, allowing for more targeted and engaging digital interventions. A recently developed mobile app intervention uses a chatbot to deliver evidence-based SUD interventions (including cognitive-behavioral therapy and motivational interviewing) in a tailored, responsive manner (Prochaska et al., 2021), which aligns with social work's emphasis on person-centered care. As technologies like this proliferate, they may help make intervention and prevention more accessible for vulnerable populations who otherwise could not access care, and they could also give social workers a rich toolkit of adjunctive treatments to enhance usual care and recovery.

## ***5.2 Real-Time Monitoring and Relapse Prevention***

Much of the promise of digital therapeutics lies in the potential to integrate rich, frequently collected data to provide timely responsive care. One source of data that could enhance these tools' usefulness is real-time data from users' smartphones and wearable devices such as smartwatches. Smartphones automatically collect data such as individuals' location and social interactions (e.g., calls and text messages), and wearable devices use biosensors to collect physiological and health data on an ongoing basis, including individuals' physical activity, cardiovascular function, and stress reactivity. Research has begun to examine how AI can use this data to identify behavioral and physiological changes that indicate a risk for relapse among individuals in recovery from SUD. One recent study demonstrated that increases in stress reactivity, measured by wearable biosensors, are associated with relapse over the following week (Mahoney et al., 2023). Another study predicted substance craving 90 min into the future with 93% accuracy solely from GPS data (Epstein et al., 2020).

Because smartphones and wearable devices continuously and automatically collect data like this, they can enable real-time monitoring of risk across multiple biopsychosocial domains without requiring user input. AI leveraging this data could power personalized interventions delivered via mobile devices that target specific risk factors in a time-sensitive manner. Such interventions, known as just-in-time adaptive interventions (Carpenter et al., 2020), could leverage smartphone and wearable device data to identify when individuals are most at risk of relapse and intervene effectively. A pilot study explored the use of this strategy with chronic pain patients at risk of problematic substance use, where participants were prompted by a mobile app to practice mindfulness when their wearable biosensors identified heightened stress reactivity (Garland et al., 2023). Practicing mindfulness in those moments was associated with reductions in pain, stress, and craving for substances.

## 6 Real-World Applications: Opioid Risk Stratification in the Veterans Affairs

Most work using AI to target health outcomes, including problematic substance use, has remained at the research stage, thus far limiting its real-world benefits. However, one early adopter has been the US Department of Veterans Affairs (VA). In the early 2010s, opioid prescriptions increased at the VA, due in part to exceedingly high rates of chronic pain among veterans (Gellad et al., 2017). Many veterans who were prescribed opioids suffered from adverse outcomes, including suicide and overdose, reflecting the broader opioid epidemic in the United States. Seeking new strategies to mitigate the risk of adverse outcomes among veterans who are prescribed opioids, the VA's extensive electronic health records and centralized data infrastructure allowed it to go beyond traditional monitoring approaches.

The VA developed and deployed the Stratification Tool for Opioid Risk Mitigation (STORM) to identify patients with an opioid prescription who are at risk of overdose or suicide (Oliva et al., 2017). STORM leverages a predictive model to calculate an up-to-date risk score for each patient, based on their demographics, prior adverse events, and physical and behavioral health history. The provider-facing STORM dashboard is a clinical decision support tool that displays the risk factors for each patient (e.g., diagnoses, specific medications) and documents whether specific risk-mitigation strategies (e.g., psychosocial treatments) have been employed with the patient. In a randomized evaluation of STORM involving over 44,000 at-risk patients at all 140 VA medical centers, use of the risk stratification tool was associated with a significant decrease in mortality over the next 4 months (Strombotne et al., 2023).

At this early stage of adoption, it is crucial to study the implementation of strategies like STORM and to glean lessons that can be applied as predictive models are deployed more broadly across health and service settings. Supportive implementation, leadership, and sufficient training are crucial. In a study of STORM's implementation in 39 VA facilities, some staff were not fully aware of its clinical importance or evidence base; there were variations in training, guidance, and leadership buy-in for STORM; and there were challenges integrating the tool into existing clinical workflows and priorities (McCarthy et al., 2022). Questions also remain about how such a tool might be applied in settings beyond the VA, especially in lower-resourced settings with a less robust data infrastructure. Still, STORM represents a pioneering example of implementing predictive analytics in substance use-related care, demonstrating how AI-powered risk stratification tools could be integrated into clinical workflows and potentially improve patient outcomes at scale.

## 7 Ethical Considerations

### 7.1 *Bias and Equity*

The rising use of AI for SUD prevention and treatment brings with it several ethical concerns, particularly for marginalized and vulnerable populations. These technologies must reduce, rather than reinforce, existing disparities in SUD and other outcomes. Machine learning models are not neutral; they inherit the biases and limitations of their training data. Many machine learning models have been developed using data from predominantly white, higher socioeconomic status populations (Turner Lee, 2018; Juhn et al., 2022). As a result, these models may perform poorly for individuals from underrepresented groups, leading to misclassification and inappropriate interventions. Machine learning can also reinforce disparities in SUD diagnoses, reflecting biases in how society defines and assigns risk. For example, models trained on criminal justice data could disproportionately flag Black and Latino individuals as high-risk due to historical patterns of racialized policing rather than actual substance use behaviors (Huq, 2019).

Beyond biased risk predictions, disparities in access to AI-driven interventions could create additional barriers to equitable care. Many of the same populations most vulnerable to SUD—low-income individuals, rural communities, and people with disabilities—also face structural barriers in accessing both treatment and digital technologies (Lutz, 2019). If AI-powered tools rely on expensive smartphones, stable Internet connections, or wearable devices, they may inadvertently exclude these groups, further widening disparities in care. Many AI-driven treatment tools are developed in well-resourced healthcare settings. They may be challenging to disseminate in lower-resourced settings or in community-based programs serving the most vulnerable. Additionally, policy constraints and funding limitations may determine whether these technologies are integrated into public healthcare systems, potentially perpetuating a two-tiered system in which socioeconomically disadvantaged individuals receive less benefit from advances in AI treatment.

### 7.2 *Social Construction of Risk*

Moving beyond bias in how AI learns from past data, we must also consider how it can actively construct social narratives of risk, shaping who society views as high-risk, what outcomes are considered problematic, and which treatment approaches are prioritized in SUD care. One issue concerns how “risk” is defined and measured in machine learning models and how this can reinforce society’s existing biases against marginalized populations (Rajkomar et al., 2018). To illustrate, a machine learning model that includes race as a possible predictor of adverse SUD outcomes and is deployed in a hospital may “learn” that racial minority patients of that hospital are more likely to develop SUD and may perpetuate an incorrect belief among

providers that minorities are doomed to have substance use problems. However, it may, in fact, be the case that living in an economically disadvantaged area is what puts patients at risk, and racial minorities are overrepresented in these areas. Suppose information on area-level economic disadvantage is not included in a predictive model. In that case, the model may learn the wrong lesson and perpetuate stereotypes about race rather than getting to the societal roots of disparities.

Similarly, we must consider how AI systems designed to detect the risk of adverse substance use outcomes automatically may reinforce unhelpful ideas about what constitutes a poor outcome. Importantly, the SUD field has seen advances in harm reduction approaches and an understanding that one size does not fit all when it comes to treatment goals. There is a risk that deploying AI models trained on a singular outcome (such as complete abstinence) will lead to less person-centered care if that outcome differs from the patient's goals. Treatment and prevention decisions driven by AI may be made to meet goals that diverge from a patient's own priorities, such as moderating their use, learning to use more safely, or pursuing specific life goals as a person in recovery.

### **7.3 *Privacy and Surveillance***

The implementation of AI systems for SUD also raises substantial privacy concerns, especially for marginalized communities who have historical reasons to distrust health and social service systems (Dovidio et al., 2008). AI applications often rely on collecting sensitive personal data, such as from electronic health records, social media, and consumer electronic devices. Communities can be wary of sharing data this broadly, and leaks or unintended disclosures involving information on substance use could lead to employment discrimination, loss of child custody, or criminal justice involvement (Looi et al., 2024). Further, when an AI system flags an individual as high-risk, this label could follow them through service systems, potentially leading to differential treatment or discrimination. This is particularly concerning for marginalized populations who are already disproportionately surveilled by social systems. For example, research shows that drug testing during pregnancy and at birth disproportionately targets low-income women and women of color, despite similar rates of substance use across demographic groups (Kunins et al., 2007; Chasnoff et al., 1990). AI could perpetuate these unfair practices.

It is critical to consider the broader potential for increased surveillance. Even when care providers have the best of intentions, relapse monitoring systems that track location or wearable device data can be perceived as intrusive (Moore et al., 2021). There is also the possibility that data and monitoring systems may be accessed by law enforcement or used in criminal proceedings. For the sake of communities that have experienced over-policing and disproportionate criminal justice involvement due to substance use, particularly Black and Latino communities, AI monitoring tools must be developed and disseminated with extreme caution (Boehme et al., 2022).

## ***7.4 Client and Provider Autonomy and Empowerment***

Though AI has the potential to empower individuals with or at risk of SUD, it could also diminish their autonomy. This is of critical concern to social workers, as respect for client autonomy and self-determination is central to social work's ethical code (NASW, 2021). On the one hand, AI systems that leverage clients' own data about their health and functioning may help them and their clinicians make more informed decisions to mitigate the risk of SUD. Wearable devices could enhance individuals' ability to self-monitor their own risk for relapse by deepening their awareness of their moment-to-moment health status. Digital therapeutics could empower individuals by providing greater access to information, support, and treatment options beyond traditional service settings.

On the other hand, AI could undermine autonomy among those with SUD risk if it is deployed unwisely or in support of over-surveillance or coercive treatments. As described above, data and monitoring tools could contribute to punitive surveillance and be used to mandate abstinence in criminal justice or employment settings. There is also a danger that over-relying on AI could diminish clinicians' ability to apply their own clinical wisdom and to have person-centered conversations with patients about their care and lead to inappropriate care decisions (Tiribelli, 2023). While AI-based treatment recommendation systems seek to empower clinicians, some models are based on difficult-to-interpret "black boxes," which might make it hard for patients and providers to understand why specific recommendations are made, to question intervention strategies critically, and to tailor decisions if necessary (Sendak et al., 2020). Suppose patients come to rely on AI to drive health decisions. In that case, they may become increasingly passive recipients of care, leaving them vulnerable to AI interventions that influence their choices without their full awareness or informed consent (Tiribelli, 2023).

## **8 Future Directions and Recommendations**

### ***8.1 Emerging Trends in AI Technology***

Next, we discuss emerging trends in AI and offer concrete recommendations for social workers as AI proliferates in SUD care. As AI rapidly advances, many next-generation technologies have significant potential to improve SUD prevention and treatment. One prominent area of research, with exciting applications for SUD, is reinforcement learning. While most machine learning models are trained on past data to make predictions, reinforcement learning continues to incorporate feedback after each decision. It adjusts its decision-making approach over time—effectively learning by doing (Shakya et al., 2023). This makes reinforcement learning especially promising for interventions that learn from ongoing user behavior and environmental changes. In SUD treatment, reinforcement learning could power

just-in-time adaptive interventions that are especially precise and effective. For example, a just-in-time intervention that uses reinforcement learning might learn, through trial and error, that a person ignores relaxation exercises but responds well to peer support. Because reinforcement learning can try an intervention and observe the results, it can dynamically learn the optimal timing of interventions and how to motivate people to engage in SUD recovery behaviors.

We are also in the midst of an explosion in the use of generative AI. In contrast to traditional AI models that classify information or make predictions, generative AI models are pretrained on vast amounts of diverse data, enabling them to flexibly understand user input and generate new content (Jo, 2023). This technology could drive increasingly interactive and responsive digital therapeutics for SUD. One key area where this could be used is in advanced chatbots, built on text-based generative AI systems known as large language models (LLMs). While earlier chatbots relied on rule-based systems with pre-scripted responses, LLM-based chatbots can engage in more natural, nuanced conversations that adapt more closely to users' specific needs and concerns. An LLM-based chatbot could deliver evidence-based counseling interventions for SUD (such as motivational interviewing or cognitive behavioral therapy) and do so in an increasingly engaging and sophisticated manner. Beyond direct client support, LLMs can assist social workers by synthesizing large volumes of clinical documentation and research. For providers managing large caseloads, LLMs could quickly summarize vast amounts of clinical data, flagging patterns that suggest risk for poor outcomes, and recommend evidence-based interventions tailored to an individual's unique needs.

## ***8.2 Promoting Harm Reduction with AI***

Much of the focus of work applying AI to problematic substance use has focused on assessing and preventing SUD, preventing relapse, and promoting abstinence. However, AI also has promise for mitigating individual and social consequences of use, in line with team-based harm reduction approaches, which are effective in addressing complex health care needs through integrated care (Leibowitz et al., 2023). AI-powered tools could support harm reduction efforts among social workers, other interdisciplinary practitioners, and policymakers. Preliminary research suggests the uses of AI to support harm reduction are broad and could include not only preventing poor outcomes like overdose in clinical settings (as in the example of the VA's STORM tool, above; Oliva et al., 2017) but also identifying larger geographic areas where risk is rising and informing optimal community resource allocation.

Research is demonstrating how AI models can help identify "hotspots" of increased drug use or overdose, potentially to optimize the distribution of preventive services. Machine learning models are being used to analyze community-level data, such as historical overdose rates, emergency medical services records, and patterns of disease transmission, to predict trends in injection drug use and overdose

(Bartholomew et al., 2022; Prieto et al., 2020). Such models help policymakers determine the most effective allocation of resources, such as safe injection sites and overdose prevention kits, and they enable community service providers to refine staffing and outreach strategies. For preventing overdose at the individual level, wearable biosensors could monitor physiological indicators of overdose, such as breathing rate and oxygen levels, and AI systems could trigger an emergency response (Ezell et al., 2024).

### **8.3 Considering Implementation Challenges and Developing Best Practices**

As AI tools for SUD treatment and prevention transition from research to real-world practice settings, their success will depend on careful attention to equity, ethical implementation, and effective integration into existing systems. Deploying and scaling AI will require significant financial and technical investment, which could pose a barrier, especially in low-resource settings where need is often greatest. Without proactive policies to promote equity, AI could reinforce disparities in SUD care rather than helping to close them. Relatedly, discrepancies in data infrastructure across settings may pose a challenge: a tool developed in one setting may not be feasible in another if it relies on resources unavailable there. AI systems that depend on comprehensive, multimodal data may not perform as well in settings where such data are unavailable, incomplete, or inconsistently collected. It will also be critical to ensure that staff with adequate AI expertise are available across systems to facilitate patients' understanding of AI-driven interventions. As many AI models function as complex "black boxes," non-expert social workers and clients may struggle to trust, critically appraise, or implement AI-generated recommendations. Finally, there is a risk of deploying AI that is out of step with social work ethics and values, such as predictive tools that promote bias and impinge on privacy, or treatment applications that diminish personal autonomy and clinical judgment.

It is therefore essential to develop best practices for developing and properly implementing AI tools in the SUD field. Given the early stage of development and rapid growth of AI in this field, the following principles are not meant to be exhaustive but to serve as a starting point for a broader conversation:

- *Multimodal, Biopsychosocial Data Integration:* AI models should incorporate a breadth of data sources to provide a more holistic view of SUD risk and treatment needs. In addition to traditional clinical data, integrating data on area-level social determinants of health, wearable device data, and other factors can help AI systems identify structural and social risks rather than reinforcing individual blame, stigma, and biases.
- *Client and Provider Autonomy and Informed Consent:* AI tools for SUD must be built to promote individual agency and informed decision-making. As these tools are deployed, informed consent procedures should be clear and ensure clients

understand how their data will be stored and used and how they can opt out of AI. The use of punitive, surveillance-based, or mandatory-participation tools should be critically questioned. Clinicians also should not offload their expertise to technology and rely on AI to make decisions for them. Instead, they should learn to critically engage with technology to augment their practice and help enhance their decision-making.

- *Explainable AI, Trustworthiness, and Transparency:* To the extent possible, AI tools should be designed for interpretability, ensuring that both practitioners and individuals receiving care can understand why and how decisions are made and that clients retain shared decision-making power. Developers should prioritize models that generate clear, human-readable explanations of risk and treatment recommendations, enabling clinicians and clients to engage with AI rather than relying on it (Xu et al., 2019). Stakeholder-driven approaches should incorporate patients' perspectives, lived experience, and domain knowledge in SUD into all phases of machine learning model development. This ensures that data (e.g., EHRs) are representative of the population and that models validly predict clinical outcomes and stratify SUD risk. This approach can also enhance interpretability, facilitate engagement, and build trust that models are appropriate for clinical use and align with clinicians' assessments.
- *Addressing Bias and Equity:* AI models should be rigorously tested across diverse populations to identify biases in risk classification and treatment recommendations. Developers must ensure that AI does not disproportionately flag marginalized groups because of systemic biases in training data and instead incorporates structural determinants of risk. Efforts must be made to ensure digital resources, including AI and data systems, are equitably distributed.
- *Personalized and Flexible, Integrated, Team-based Treatment:* As AI systems can consider many biopsychosocial risk and protective factors, research evidence, and clients' preferences concurrently in real time, they could support a revolution in personalized and responsive care for SUD. To this end, AI-driven interventions should adapt to individual goals, needs, and treatment preferences, rather than adopting a one-size-fits-all approach. In line with integrated hub-and-spoke approaches and recognizing that abstinence-only models may exclude many individuals with different goals, AI tools should support harm reduction and recovery capital (Best & Hennessy, 2022). Social workers in both hubs and spokes serve on MAT teams alongside medical providers, promoting integrated behavioral health and interdisciplinary collaboration and linking clients to social services. Similarly, they can facilitate telehealth and digital integration.
- *Inclusive Design and Co-creation:* AI for SUD should be developed in partnership with people with lived experience and those in recovery—especially individuals with intersecting, marginalized identities—as well as SUD service providers and community organizations. AI tools should align with these individuals' real-world needs, preferences, and ethical concerns. Digital therapeutics and clinical decision support tools should be user-friendly, engaging, culturally relevant, interpretable, and intuitive to use to increase engagement, minimize adoption barriers, and reinforce equitable access to care.

## ***8.4 Implications for Social Work Education and Practice***

AI has the potential to improve SUD prevention and treatment by improving risk assessment, expanding access to services, and supporting targeted biopsychosocial interventions. Given potential ethical challenges and implementation barriers, social workers should have a seat at the table as these technologies are being developed and deployed, and they can help give voice to the needs and concerns of vulnerable populations with SUD. In doing so, social workers can help ensure AI is implemented in a way that supports rather than impinges on health and social justice.

One of the most pressing concerns is the need for enhanced data and technology literacy in social work education (Rodriguez et al., 2024; Creswell Báez et al., 2025). AI tools may increasingly shape clinical decision-making and policy decisions for people at risk of SUD. Yet, social work professional education typically includes little to no formal training on how these systems operate. To effectively inform clients and advocate for ethical AI use, social workers must be equipped with foundational knowledge of AI, predictive modeling, and digital equity. Schools of social work should update their curricula to ensure that graduates are prepared to navigate and critique advances in AI as they relate to practice and policy in SUD and related fields. Practicing social workers should pursue continuing education in this area to stay up to date on these rapidly advancing technologies.

Beyond education, social workers can play an active role in ensuring that AI serves marginalized communities with SUD, rather than reinforcing disadvantages. When working with clients, we must be prepared to help them understand how AI tools collect and use sensitive data, as well as the potential benefits, risks, and shortcomings of these systems. We should also educate our clients on how their data are used, privacy shortcomings and safeguards, and their rights to opt out of specific services or features. As program managers and advocates, we should promote equitable and ethical AI and push for policies that support transparency, mitigate bias, and lower barriers to technology among underserved communities.

Social workers should work closely with stakeholders and technical experts, including data scientists, software developers, and program managers, to ensure that these tools are designed with an accurate grasp of SUD and its etiology. That is, we should advocate for AI tools that align with a biopsychosocial view of SUD and address clients' real-world needs, rather than taking a one-size-fits-all approach. Finally, we should interrogate how AI in legal, medical, and social service settings can reinforce stigma and criminalization among marginalized populations, and we should prioritize empowering and harm reduction-based approaches.

## 9 Conclusion

In summary, this chapter provides a comprehensive exploration of the current state of AI in SUD treatment and prevention. Advances in AI and related data infrastructures have enormous transformative potential. AI-based tools can match the multifaceted and complex nature of SUD, democratize information and care, more accurately model risk, and make treatment and prevention more personalized, timely, and effective. Yet, these advances must be grounded in social work values to ensure that AI implementation is ethical, equitable, and does not reinforce existing biases and disparities. This chapter begins a conversation on best practices for AI in SUD and considers implications for social work education and practice.

As AI moves beyond controlled research studies into SUD practice settings, we call on social workers to engage with data scientists, technologists, and policymakers and collaborate closely on the development and implementation of these tools. Our expertise can help ensure that AI-based predictive models and digital therapeutics are attuned to the full complexity of SUD as a biopsychosocial problem and are grounded in the real-world challenges and concerns of vulnerable populations. As professionals who routinely interface with and advocate for marginalized communities at risk of SUD, we should view our role primarily as giving voice to these communities and work to ensure that technological advances are not only effective but also balanced with humanity and social justice.

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